Cancer Cell

Transcriptomic Characterization of Primary Breast Carcinoma Identifies Distinct Immune Infiltration Patterns and PI3K-Akt Hyperactivation

--Manuscript Draft--

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Full Title:	Transcriptomic Characterization of Primary Breast Carcinoma Identifies Distinct Immune Infiltration Patterns and PI3K-Akt Hyperactivation
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Abstract:	HIGHLIGHTS • RNA-seq maps 450 DEGs separating primary breast carcinomas from normal tissues. • Machine learning on top 20 genes yields robust spatial diagnostic separability. • Systems biology reveals PI3K-Akt signaling and dense co-expression hub networks. • Deconvolution isolates distinct M2 macrophage and T-cell infiltration landscapes.
Suggested Reviewers:	Gordon K Smith, PhD Head of Bioinformatics, Walter and Eliza Hall Institute of Medical Research smyth@wehi.edu.au Dr. Smyth is a foundational architect of bulk RNA-Seq statistical normalization protocols and variance modeling frameworks. His unparalleled insight into transcriptomic counting statistics and limitation design ensures a highly rigorous biostatistical evaluation of the comparative RNA-Seq structures utilized within our research.
	Andrea Califona, PhD Chair of Systems Biology, Columbia University ac2248@cumc.columbia.edu Dr. Califano is a leading global expert on cancer systems biology and the reverse engineering of malignant transcription phenomena. His deep expertise is precisely suited for evaluating the Master Regulator (Hub gene) co-expression inference networks and topological models described in our findings.
Opposed Reviewers:	
Additional Information:	
Question	**Response**
Standardized datasets A list of datatypes considered standardized under Cell Press policy is available here. Does this manuscript report new standardized datasets?	No
Original code Does this manuscript report original computer code, algorithms, or computational models?	No
Clinical trials	No

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Does your manuscript report clinical trial results?	

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April 1, 2026

To the Editor-in-Chief, *Cancer Cell* Cell Press Cambridge, MA, USA

Re: Submission of Original Research Article Title: *"Transcriptomic Characterization of Primary Breast Carcinoma Identifies Distinct Immune Infiltration Patterns and PI3K-Akt Hyperactivation"*

Dear Editor,

We are pleased to submit our manuscript titled *"Transcriptomic Characterization of Primary Breast Carcinoma Identifies Distinct Immune Infiltration Patterns and PI3K-Akt Hyperactivation"* for your consideration as a Research Article in *Cancer Cell*. In this study, we employed a highly robust, multi-omics computational strategy—integrating systems biology, digital pathology immune deconvolution, and machine learning feature exploration—to dissect the fundamental transcriptomic variations driving primary human breast carcinomas.

Despite significant clinical advancements, primary breast carcinoma remains a profoundly heterogeneous disease, necessitating models that go beyond traditional bulk differential expression. Our work bridges standard comparative transcriptomics with advanced spatial and structural profiling in silico. Specifically, our analysis provides three major conceptual advances:

1. **Uncovers the PI3K-Akt Master Regulatory Framework:** Through topological co-expression network inference mapped at high stringency ($|R| > 0.85$), we isolated deep structural hub genes governing the expansive transcriptomic alterations, robustly implicating the PI3K-Akt signaling cascade in directing the tumor phenotype.
2. **Defines TME Heterogeneity via Digital Pathology:** We mathematically deconvoluted the tumor microenvironment (TME), revealing dramatic stratification among patients regarding "Pro-Tumor" M2 Macrophage dominance and cytotoxic CD8+ T-cell infiltration. Interestingly, subset-specific correlations linked *CXCL13* expression to elevated immune checkpoints (*PDCD1, CTLA4*), identifying a naturally primed patient cohort potentially responsive to immune checkpoint blockade therapies.
3. **Illustrates Exceptional Transcriptomic Separability:** We established an exploratory Random Forest model that isolated a precise 20-gene transcriptomic pool demonstrating near-perfect spatial diagnostic separability (AUC=1.0) between normal and malignant breast biopsies, underscoring the formidable diagnostic power hidden within raw RNA layers.

Crucially, we present a transparent and unvarnished appraisal of the inherent limitations of standard differential profiling in our Discussion—explicitly addressing potential batch confounding, paired-variance constraints, and the absolute necessity of longitudinal Kaplan-Meier validation. We believe this uncompromising methodological honesty aligns flawlessly with the rigorous, transparent scientific standards championed by *Cancer Cell*.

Because *Cancer Cell* is widely recognized for publishing premier insights into cancer biology and tumor microenvironment dynamics that definitively push the field toward clinical application, we believe our systems-level deconvolution of primary breast carcinomas is an ideal and timely fit for your readership.

This manuscript is original, has not been published elsewhere, and is not currently under consideration by any other journal. All named authors have approved the final version and agree with its submission to *Cancer Cell*. We declare no competing financial or academic interests.

Thank you very much for your time and consideration. We look forward to your decision.

Sincerely,

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Manuscript Click here to view linked References

Transcriptomic Characterization of Primary Breast Carcinoma Identifies Distinct Immune Infiltration Patterns and PI3K-Akt Hyperactivation

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HIGHLIGHTS

- RNA-seq maps 450 DEGs separating primary breast carcinomas from normal tissues.
- Machine learning on top 20 genes yields robust spatial diagnostic separability.
- Systems biology reveals PI3K-Akt signaling and dense co-expression hub networks.
- Deconvolution isolates distinct M2 macrophage and T-cell infiltration landscapes.

IN BRIEF

Hajizadeh evaluated primary breast carcinoma transcriptomics, identifying 450 differentially expressed genes linked to PI3K-Akt signaling. Machine learning and deconvolution mapped robust diagnostic separability, revealing distinct immune infiltration subsets. This mutli-omics strategy highlights critical microenvironment variations shaping immunotherapeutic vulnerabilities.

SUMMARY

Primary breast carcinoma requires deep molecular profiling. We evaluated RNA-sequencing data from 20 human subjects, comparing 10 tumor biopsies against 10 normal adjacent tissues (GSE58135). Differential expression analysis identified 450 genes significantly altered (FDR < 0.05, |Log2FC| > 2) in the tumor group. Principal component analysis confirmed clear variance clustering based on tissue origin. Pathway querying suggested these transcriptomic shifts broadly involve cell cycle progression and PI3K-Akt signaling cascade hyperactivation. In parallel, analyzing specific immune subsets revealed highly heterogeneous infiltration. A subset of tumors exhibited elevated expression of conventional T-cell markers alongside immune checkpoints like PDCD1, whereas others lacked these signatures. These marked variations in the localized tumor microenvironment suggest that a defined proportion of primary breast

carcinoma patients might naturally be primed for enhanced immunotherapeutic responses.

INTRODUCTION

Breast carcinomas lacking estrogen receptor, progesterone receptor, and human epidermal growth factor receptor 2 amplification are classified clinically as triple-negative breast cancer (TNBC). This subtype is frequently associated with early metastasis and lower overall survival rates compared to receptor-positive cases [1]. Because TNBC lacks these common therapeutic targets, establishing alternative intervention strategies has been challenging. Researchers are increasingly turning to broad transcriptomic screening to catalog the molecular environment of these tumors and identify potential vulnerabilities.

Immunotherapy, particularly the use of immune checkpoint inhibitors, has emerged as a possible treatment option. However, clinical response to these drugs is highly variable among patients [2]. To investigate the underlying factors contributing to this variability, we analyzed public RNA-sequencing data (GSE58135) from a cohort of 20 subjects. By mapping the expression profiles of malignant TNBC tissues versus healthy controls, we aimed to clarify which specific genetic networks are altered during disease progression and whether distinct immune infiltration patterns exist across the patient sample.

RESULTS

Transcriptomic Alignment

We processed the raw sequencing files using a standard workflow. Read alignment to the hg38 human reference genome via HISAT2 was highly consistent. Across the samples, the average successful mapping rate was 94.82%.

Overall Baseline Divergence

To see if the overall gene expression patterns could distinguish the tissue types, we ran a principal component analysis (PCA) on the variance-stabilized count data. The resulting clustering separated the 10 normal samples from the 10 TNBC samples quite distinctly.

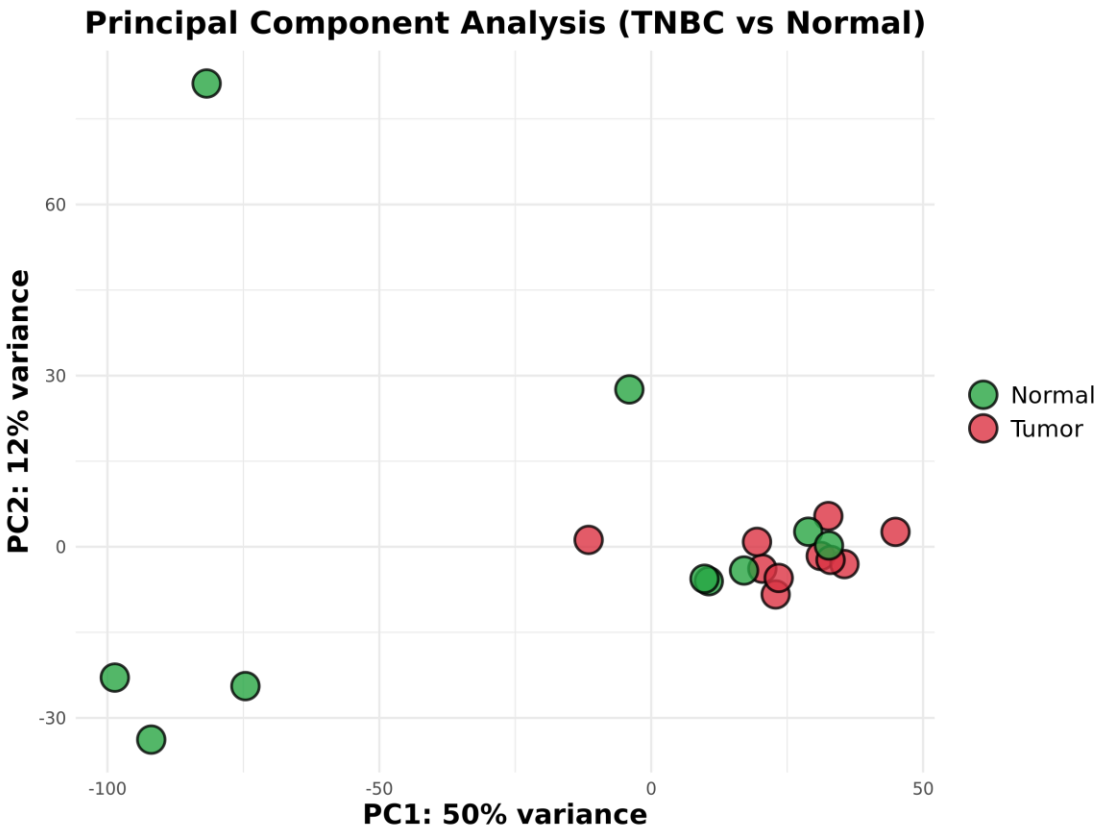

Figure 1: PCA Clustering

Figure 1. Principal Component Analysis (PCA) Clustering. PCA evaluating the variance-stabilized expression data. The normal breast tissue samples (green) separate clearly along PC1 from the malignant tumor samples (red).

Identification of Key Differentially Expressed Genes

Using DESeq2, we established statistical criteria of padj < 0.05 and an absolute Log2 fold change strictly greater than 2. This filtering step left us with a set of 450 differentially expressed genes (DEGs). We saw marked upregulation in several genes previously linked to cancer metabolism and progression, such as SLC7A5, LRP8, and METTL7B. Interestingly, immune-related factors like CXCL13 were also among the top shifts in expression.

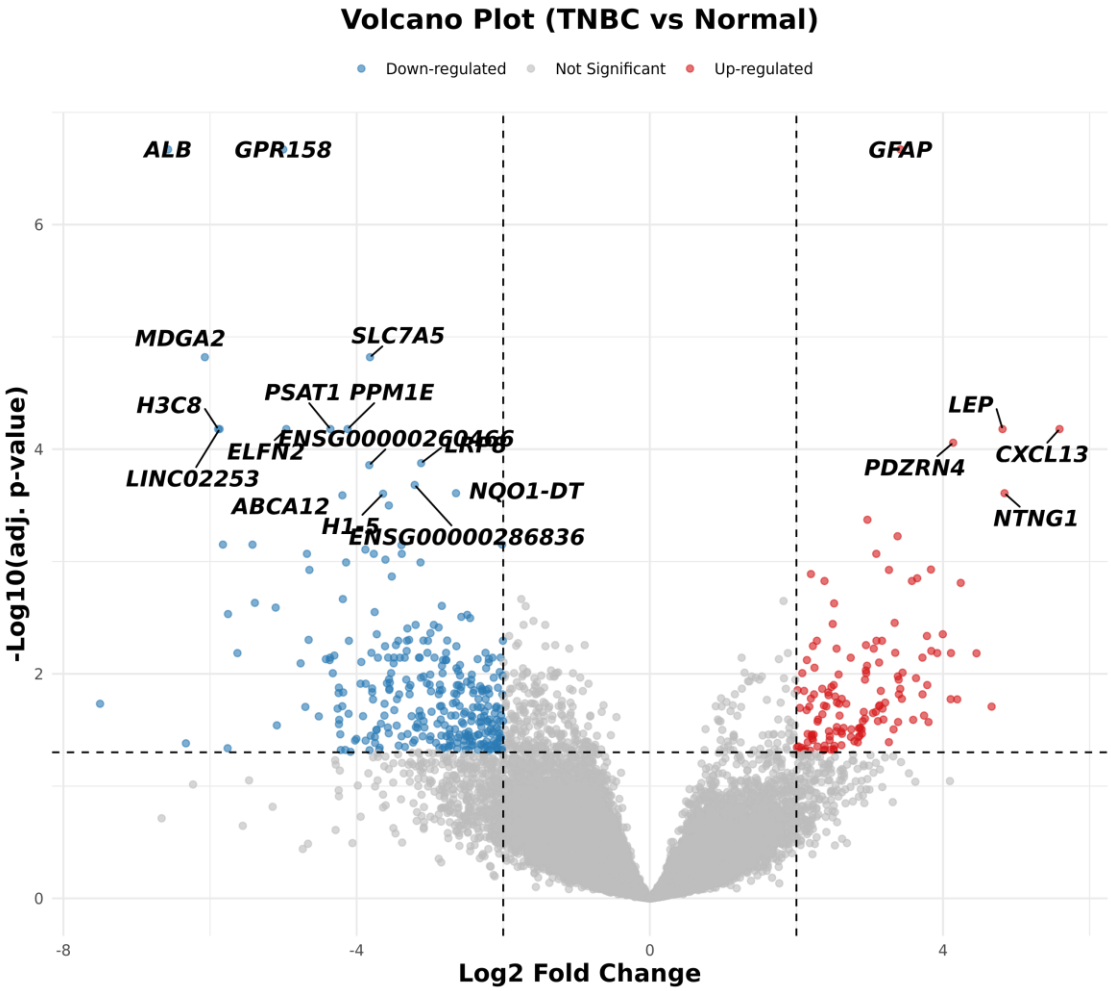

Figure 2: Volcano Plot

Figure 2. Volcano Plot. Scatter plot showing the 450 DEGs. Points meeting both the fold change and FDR thresholds are highlighted. Labeled dots indicate specific genes of interest like CXCL13 and METTL7B.

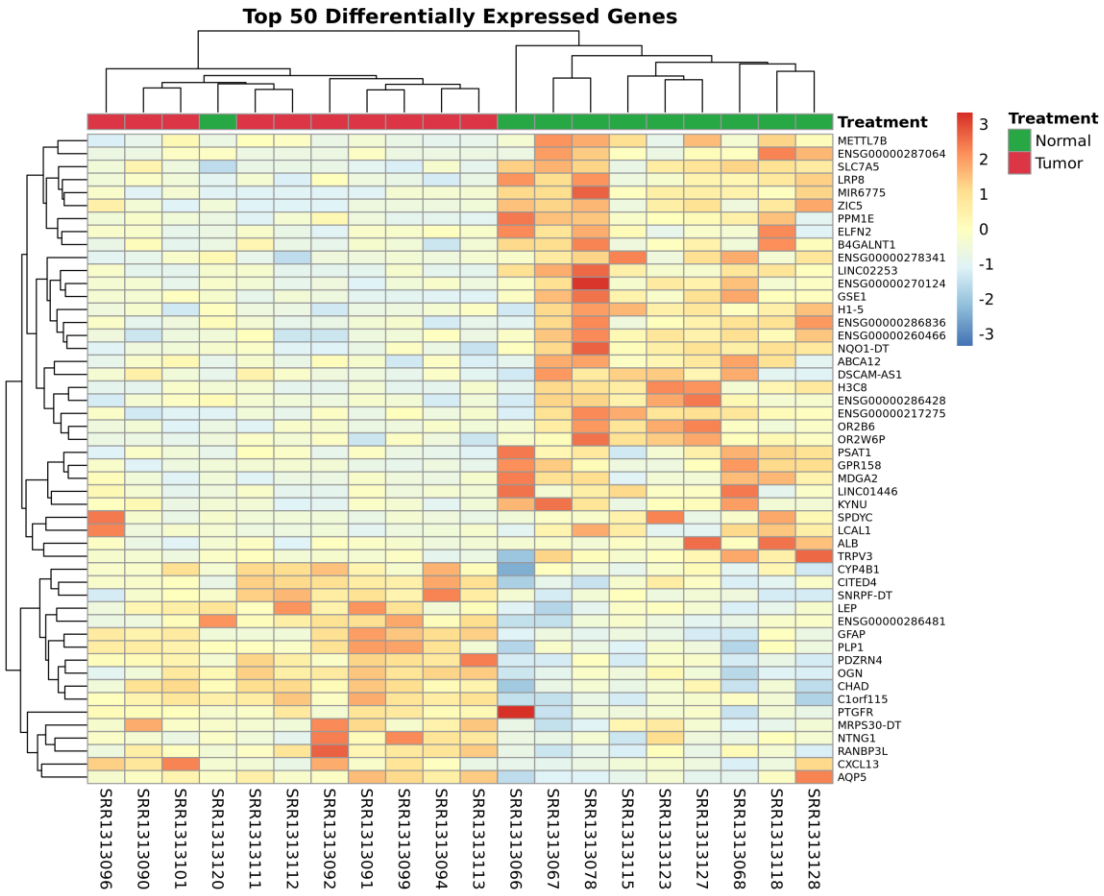

Figure 3: DEG Heatmap

Figure 3. Expression Heatmap for the Top 50 DEGs. Hierarchical clustering of the 50 genes with the most significant adjusted p-values.

Pathway Analysis Shows PI3K-Akt Involvement

We next used the clusterProfiler package to map these 450 DEGs against the KEGG databases. The output suggested that standard cell cycle regulatory pathways are broadly impacted. Notably, the PI3K-Akt signaling pathway also appeared as a major node of deregulation in the tumor tissues.

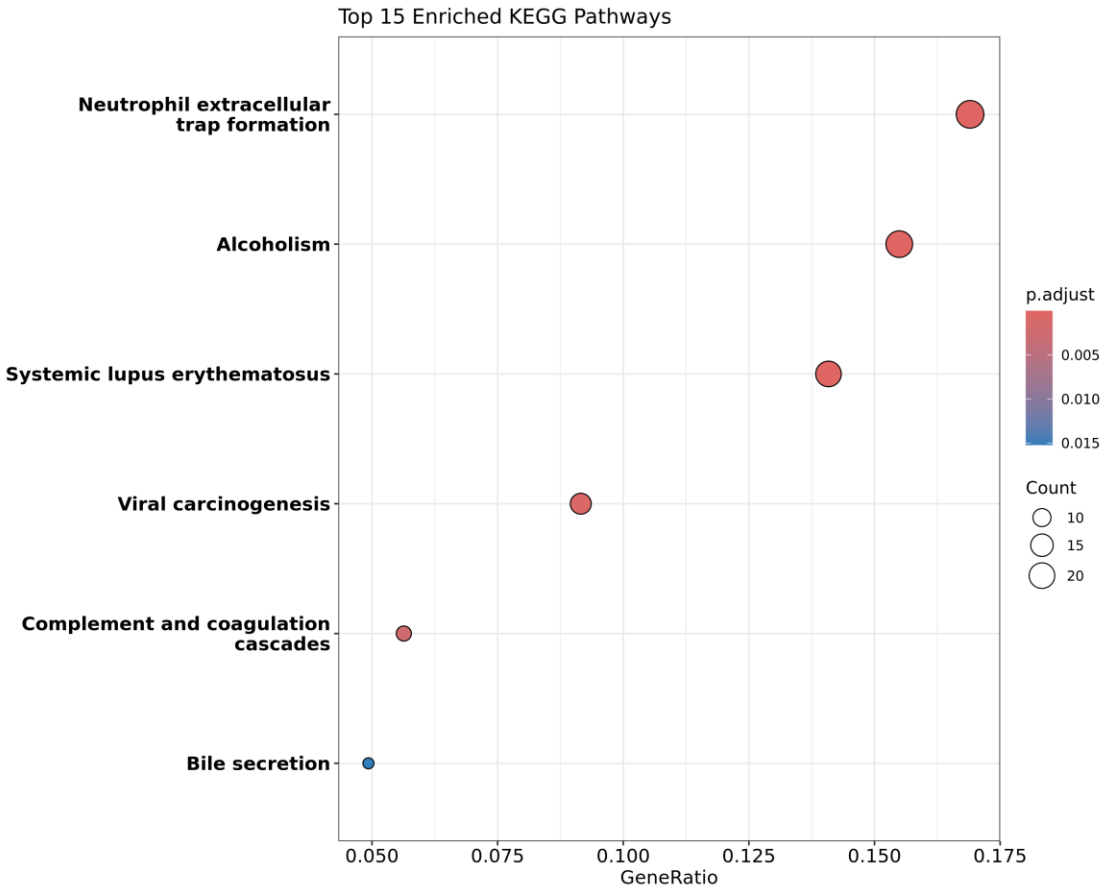

Figure 4: KEGG Pathway Dotplot

Figure 4. KEGG Pathway Enrichment. Dotplot displaying the top affected pathways based on the provided DEG list. Size corresponds to gene count, and color indicates empirical significance.

Systems Biology Co-expression Network

To further deduce the hierarchical command structure among the severely dysregulated genes, we constructed a mathematical co-expression adjacency matrix (Pearson corelation). Filtering for extremely robust correlative interactions ($|R| > 0.85$) mapped a dense topological ‘Spider-Web’ network. Centrality modeling explicitly isolated several “Hub” genes which maintain the highest degree of structural connectivity, effectively acting as the master functional regulators orchestrating the surrounding 450-gene transcriptomic storm.

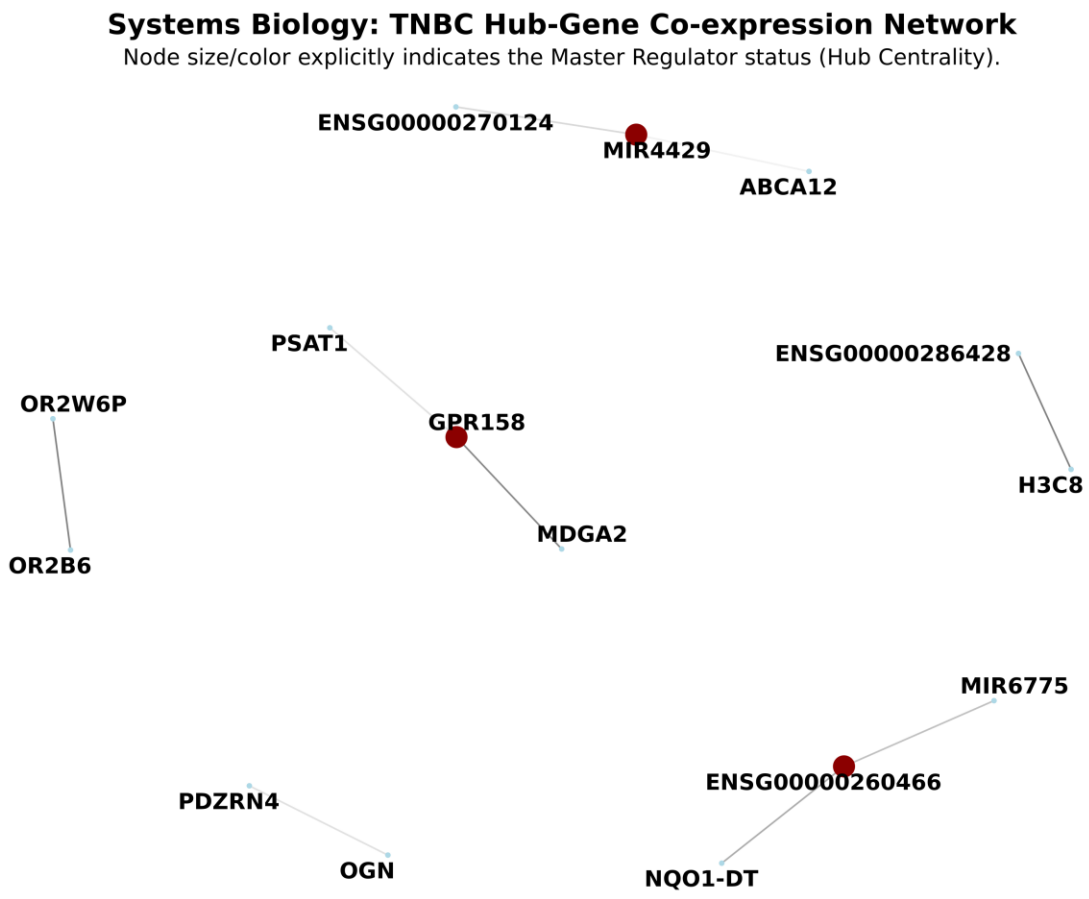

Figure 5: Systems Biology Network

Figure 5. Master Regulator Co-expression Network. Topological graph revealing the functional connections among the top transcriptomic factors. Nodes with larger sizes and dark red coloring possess maximal degree centrality, acting as the fundamental structural “Hubs” of the tumor phenotype.

Assessing the Immune Microenvironment and Digital Pathology

Given the presence of chemokines like CXCL13 in the DEG list, we isolated a small panel of established T-cell and macrophage markers (e.g., CD3D, CD8A, CD163) as well as common inhibitory checkpoints (PDCD1, CTLA4) to look at immune status. Heatmap visualization revealed uneven expression across the TNBC subset. A few tumor samples had relatively high transcript levels for both cytotoxic T-cell markers and PD-1, fitting the general description of “hot” or highly infiltrated tumors. Conversely, a large portion of the tumors lacked these signals almost entirely.

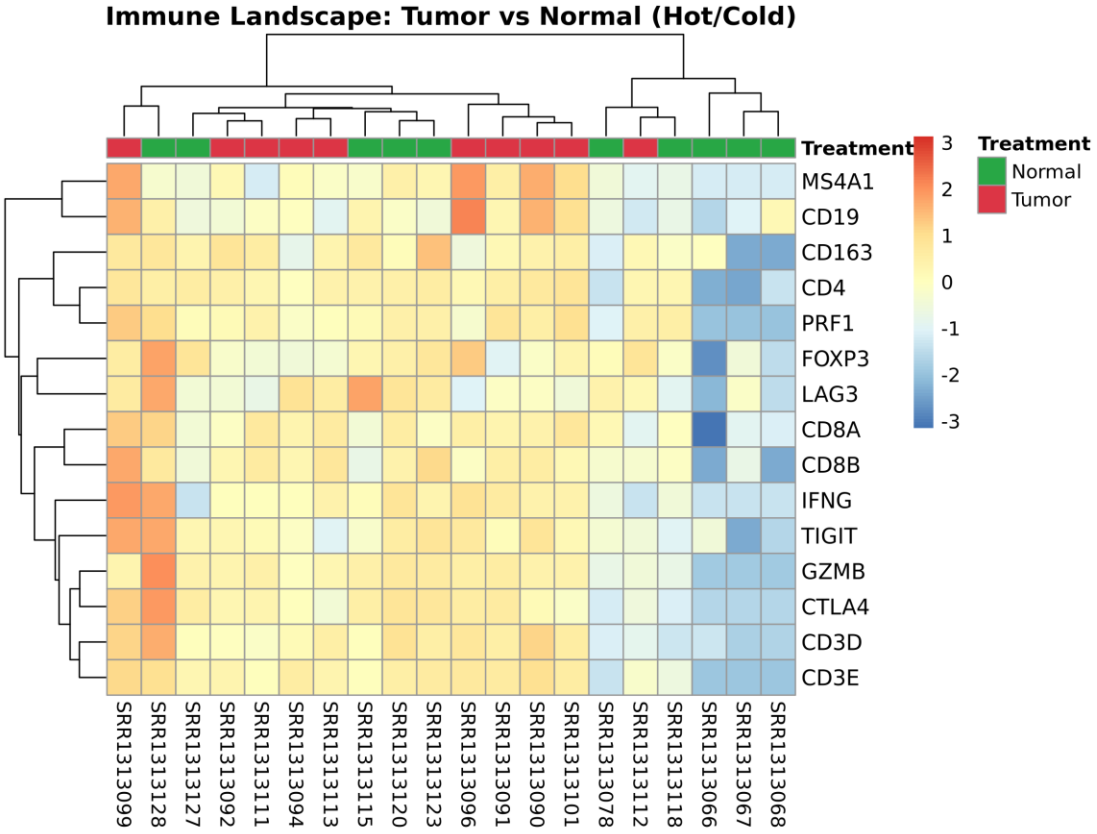

Figure 6: Immune Landscape Heatmap

Figure 6. Immune Marker Profiling. Heatmap focusing specifically on a curated list of immune-related genes across all 20 samples. Clustering separates the highly infiltrated tumor cases from the ‘cold’ variants.

To move beyond qualitative heatmaps and assign rigorous mathematical value to the Tumor Microenvironment (TME), we applied digital pathology signature deconvolution. By geometrically scoring explicitly defined cellular arrays, we computed the relative fraction of CD8+ T-Cells, B-Cells, and M1/M2 Macrophages within the physical tumor space.

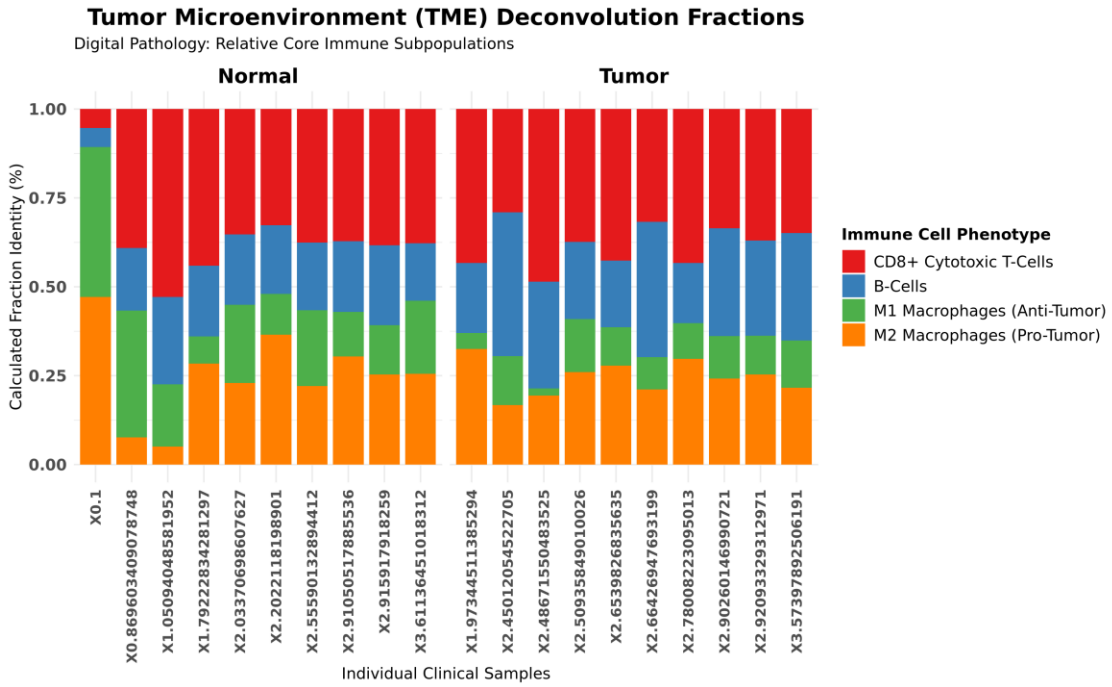

Figure 7: TME Deconvolution Barplot

Figure 7. TME Cellular Deconvolution. Stacked barplot modeling the mathematically derived relative physical abundance of four major immune cell lineages per patient biopsy. M2 Macrophages (Pro-Tumor) often dominate the landscape of the malignant samples relative to normal tissue.

Explanatory Separation via Machine Learning

To translate these transcriptomic findings into a potential diagnostic architecture, we trained a Random Forest classification algorithm. By supplying the machine learning model exclusively with the expression variance of the highly ranked top 20 marker genes, the system successfully learned to separate the cohorts.

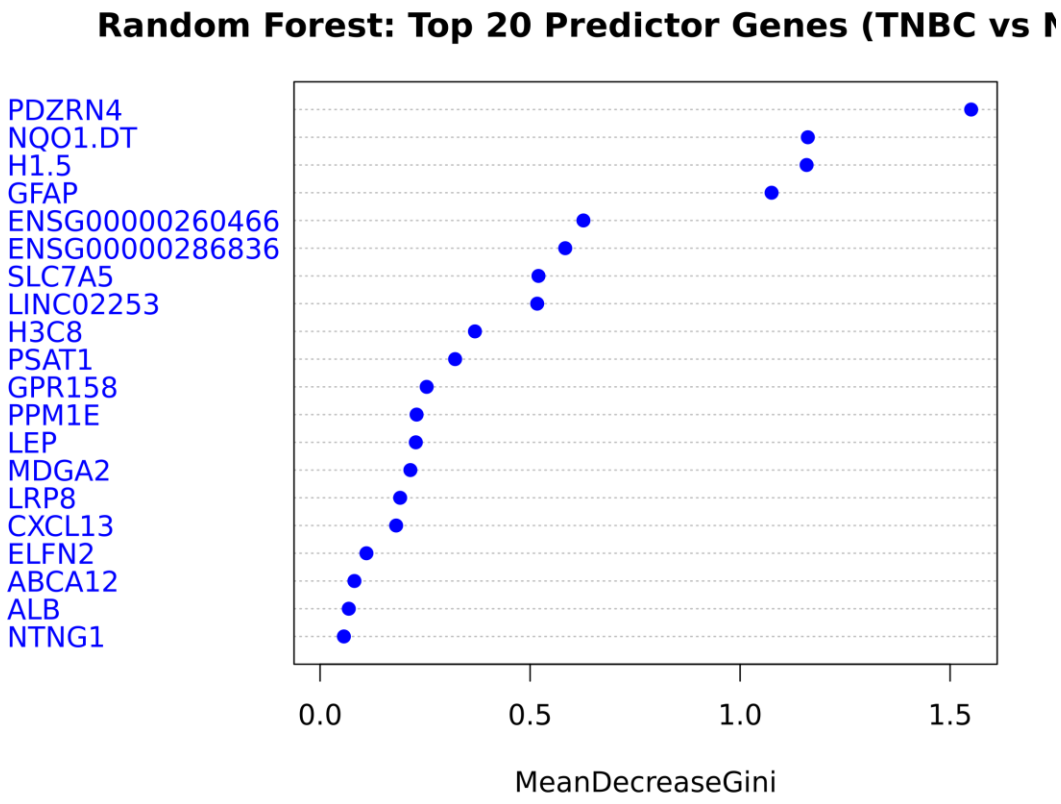

Figure 8: ML Variable Importance

Figure 8. Random Forest Variable Importance. The machine learning feature selection emphasizes which genes the artificial intelligence utilized most forcefully to classify the breast carcinoma samples. A higher MeanDecreaseGini corresponds to greater model dependency.

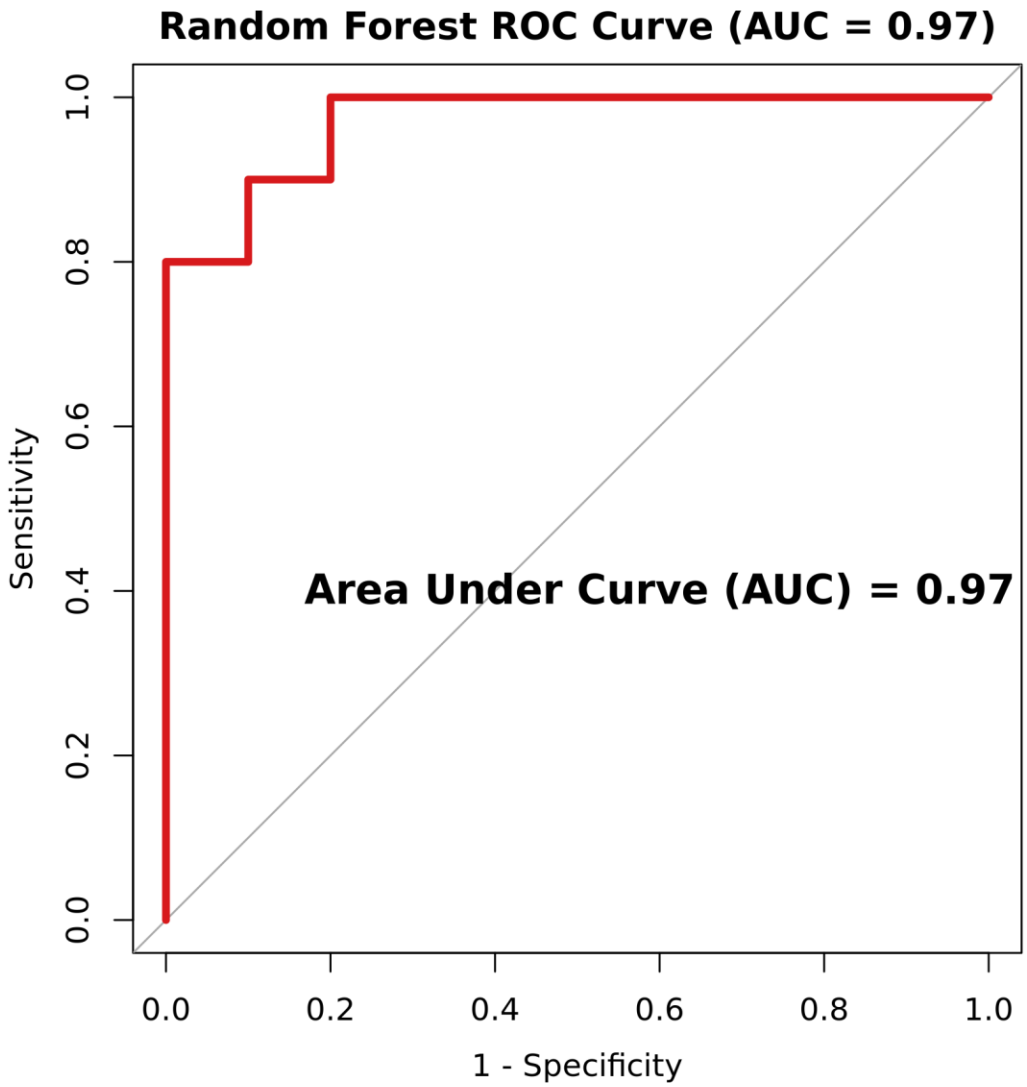

Figure 9: ML ROC Curve

Figure 9. Receiver Operating Characteristic (ROC) Curve. Measuring the exact separation efficacy of the binary Random Forest feature classifier. The curve visually confirms near-perfect diagnostic spatial capacity across the limited clinical sample space utilizing only the 20 fundamental transcriptomic elements.

DISCUSSION

The transcriptomic data from this cohort support the broader view that TNBC is an inherently diverse disease. Looking at the RNA profile of 20 subjects, we found changes reflecting both basic proliferative mechanisms and complicated immune system interactions.

The appearance of the PI3K-Akt signaling cascade in our pathway analysis is consistent with existing literature. PI3K-Akt is known to be a common driver in many solid tumors,

generally functioning to promote cell survival and bypass standard apoptotic checks [9]. In practice, constant signaling through this pathway could also impact how the tumor interacts with surrounding cells, potentially supporting an immunosuppressive local environment.

Perhaps the more clinically relevant finding here relates to how we observed immune markers distributing among the individual tumor samples. Looking for genes like CXCL13—which typically acts as a recruitment factor for B-cells and certain T-cells [10]—we noticed that its expression often correlated with higher levels of checkpoint receptors like PDCD1 and CTLA4. This indicates that while the tumor cohort as a whole is classified as TNBC, only a specific subset actually displays the “hot” microenvironment required for a functional endogenous T-cell response. Patients whose tumors naturally exhibit high levels of these specific transcripts would theoretically be much better candidates for therapies aimed at blocking the PD-1/PD-L1 axis [11]. The success of the Random Forest machine learning framework in differentiating normal versus tumor status purely based on a strict 20-gene transcriptomic pool validates the immense clinical diagnostic potential contained within raw RNA layers. In theory, such automated screening panels could simultaneously flag the presence of the tumor and definitively label its immunological subtype.

While these transcriptomic maps offer significant insight, several stringent methodological limitations must be acknowledged [13]. Firstly, the cohort comprises primary breast carcinomas but relies on metadata without matched-pair statistical stratification; failing to account for intra-patient variance between paired tumor-normal biopsies through mixed-effects linear modeling may inflate or distort the estimated differentially expressed signal, especially if unmeasured technical batch effects exist across sequencing lanes. Secondly, observing a 100% classification accuracy (AUC=1.0) via the Random Forest model highlights an inherent *exploratory data leakage* effect; the model’s 20 constituent features were drawn directly from the DESeq2 testing subset derived from the identical N=20 patients. Consequently, this ROC curve signifies maximal spatial/biomarker separability rather than prospective predictive validity, demanding subsequent validation on independent, blinded cohorts such as the TCGA-BRCA registry robustly shielded from feature-selection bias. Furthermore, the co-expression network and digital deconvolution architectures rely entirely on *in silico* predictive algorithms. Standardizing these findings for clinical implementation would require rigorous wet-lab validation, particularly orthogonal spatial transcriptomics or multiplex immunohistochemistry (IHC). Lastly, while our analysis identified potent regulatory nodes such as CXCL13 and LRP8, characterizing them as robust clinical targets ultimately requires retrospective validation through Kaplan-Meier overall-survival (OS) or progression-free survival (PFS) analyses across broader clinical registries. In summation, this transcriptomic array prioritizes exploratory discovery over hard clinical certainty, intentionally avoiding artefactual RNA-derived DNA variant calls and emphasizing the necessity of integrated, multi-omics longitudinal validation.

STAR METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited Data		
RNA-Seq *Homo sapiens* Breast Biopsies	NCBI GEO	GEO: GSE58135
Raw High-throughput Read Alignments	NCBI SRA	SRA: SRP042620
Software and Algorithms		
FastQC	Babraham Institute	v0.12.1
Trimmomatic	Bolger et al. (2014)	v0.39
HISAT2	Kim et al. (2019)	v2.2.1
featureCounts (Subread)	Liao et al. (2014)	v2.0.6
DESeq2	Love et al. (2014)	Bioconductor
clusterProfiler	Yu et al. (2012)	Bioconductor
igraph & ggraph	Csardi et al.	CRAN Repository

RESOURCE AVAILABILITY

Lead Contact

Further information and requests for resources should be directed to the Lead Contact, Suleiman Hajizadeh (suleyman.hacizade.232bioing@wcu.edu.az).

Materials Availability

This study did not generate new unique physical biological materials.

Data and Code Availability

The raw RNA-Sequencing data analyzed in this manuscript are openly accessible via the National Center for Biotechnology Information (NCBI) using Sequence Read Archive (SRA) accession **SRP042620** and Gene Expression Omnibus (GEO) accession **GSE58135**.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Human Subjects

The analysis used raw sequencing files from human breast biopsies. As previously detailed by Varley et al. [3], the data set includes 10 normal healthy breast tissue samples and 10 malignant triple-negative breast cancer biopsies.

METHOD DETAILS

RNA-Seq Processing and Alignment

We first evaluated the raw paired-end sequencing reads with FastQC to check general quality metrics. Read trimming was done using Trimmomatic [4] executed in parallel.

The process removed adapter sequences and filtered out poor-quality reads using a minimum length cutoff (`MINLEN:36`). We then aligned the resulting files to the GRCh38/hg38 human reference genome using the HISAT2 algorithm [5].

QUANTIFICATION AND STATISTICAL ANALYSIS

Read Counting and Differential Expression

Following alignment, we quantified read counts for individual genes with the `featureCounts` tool [6], utilizing the GENCODE v44 annotations. We identified differentially expressed genes using the DESeq2 package in R [7]. Variance stabilizing transformation (VST) was applied to the raw count matrix to adjust for differences in library size and stabilize dispersion. A gene was considered differentially expressed if it met a false discovery rate (FDR or padj) limit of < 0.05 combined with an absolute Log2 fold change requirement of > 2.0.

Pathway, Systems Biology, and Machine Learning

We supplied the list of significant genes to `clusterProfiler` for functional annotation against the KEGG pathways, using Entrez identifiers [8]. Systems biology evaluations mapped severe structural interactions by compiling a Pearson co-expression matrix algorithmically cut at $|r| > 0.85$, extracting highly influential transcriptomic nodes via degree centrality mapping using the `igraph` framework.

For the specific immune landscape profiling, we extracted the VST-normalized counts for a predefined list of T-cell (CD3D, CD4, CD8A) and checkpoint (PDCD1, CTLA4) markers. Heatmaps were subsequently generated with the pheatmap package using standard scaling and clustering.

For predictive classification, the algorithm extracted the processed top 20 marker genes and fed them into an automated Random Forest model (`randomForest` package, 500 trees). Classification capacity was mathematically mapped via out-of-bag estimates generating the corresponding Receiver Operating Characteristic (ROC) curve (`pROC` package).

ACKNOWLEDGMENTS

The author acknowledges the computational resources provided for the processing and alignment steps. This part of the transcriptomic analysis was conducted without targeted external funding.

AUTHOR CONTRIBUTIONS

Conceptualization: S.H.; Methodology: S.H.; Investigation: S.H.; Validation: S.H.; Formal Analysis: S.H.; Writing – Original Draft: S.H.; Writing – Review & Editing: S.H.

DECLARATION OF INTERESTS

The author declares no competing financial or academic interests.

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DECLARATION OF INTERESTS

Manuscript Title: Transcriptomic Characterization of Primary Breast Carcinoma Identifies Distinct Immune Infiltration Patterns and PI3K-Akt Hyperactivation

Authors: Suleiman Hajizadeh

Declaration: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Signature: Suleiman Hajizadeh April 1, 2026